

Hoang Vu

647-540-7166 | Email | LinkedIn | Github | Portfolio

EDUCATION

York University | GPA 7.71/9

Honours Bachelor of Science in Computer Science

Sep 2024 – Apr 2026

Toronto, ON

Sheridan College | GPA 3.78/4

Advanced Diploma in Software Development and Network Engineering

Sep 2020 – Dec 2023

Oakville, ON

PROJECTS

MindForge | *FastAPI, React, TypeScript, Supabase, OpenAI, Groq, Hugging Face, Ollama Cloud*

- Built a multi agent orchestration for micro learning platform with retry logic and fallback routing, **reducing content generation costs by 80%**
- Automated ingestion and transformation of user learning data to personalize lesson and quiz generation within 3 minutes

Academic Scheduler | *FastAPI, React, TypeScript, Supabase, OpenAI, PaddlePaddle, LightGBM*

- Developed an end-to-end data pipeline ingesting unstructured syllabus PDFs via PaddleOCR, transforming and loading structured schedule data into PostgreSQL
- Engineered features and trained a LightGBM model to predict student workload distribution, **reducing deadline clustering by 40% across 50 student schedules**

EXPERIENCE

Software Developer

SIRT Centre

May 2023 – Aug 2023

Toronto, ON

- Designed and implemented multiplayer menu system in C++ featuring character selection, persistent profile management and real-time session state tracking
- Diagnosed and resolved race conditions through thread-safety analysis and unit testing, improving system reliability for **10+ concurrent users**

Software Developer

Cloud DX

May 2022 – Dec 2022

Oakville, ON

- Built voice controlled interaction system for HoloLens2, showcased at **ANA Avatar XPRIIZE 2022** with live demonstrations
- Debugged data transmission issue between Arduino sensors and HoloLens2, tracing the root cause through C++/C# communication layer and restoring accurate sensor display
- Implemented gesture-based interaction for HoloLens2, improving task completion speed by **20%** through usability testing

Software Developer

Medtech Canada

Jan 2022 – Apr 2022

Oakville, ON

- Migrated deprecated VR architecture from Unity XR Rig to XR Origin, resulting in **40% faster scene load times**
- Resolved a critical rendering bug within **48 hours**, restoring accurate motion-tracking data

TECHNICAL SKILLS

Languages: Java, C#, Python, C++, TypeScript, SQL

Frontend: React, Bootstrap, jQuery

Backend: FastAPI, Flask, Spring Boot, ASP.NET, Node, Express

AI/ML: pandas, NumPy, scikit-learn, LightGBM, PaddleOCR

Database: PostgreSQL, MongoDB, Supabase

Tools: Git, VS Code, Visual Studio, Eclipse